MILITARY SPECIFICATION

INSTRUMENTS, SIMULATED, AIRCRAFT TRAINING DEVICES; TORQUE PRESSURE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The technical requirements for procuring the instrument described herein shall consist of this document and the issue in effect of Specification MIL-I-82356.

1. SCOPE

1.1 This specification covers the detail requirements for simulated aircraft torque pressure indicators for use in aircraft training devices.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-7304	Indicator, Pressure, Torque, Sensitive, Type A-9
MIL-I-7307	Indicator, Pressure, Torque, Type A-10
MIL-S-20708	Synchros, General Specification For
MIL-I-82356	Instruments, Simulated, For Aircraft, Training Devices, General Specification For

STANDARDS

Military

MS 27572	Case, Instrument, Flange-Mounted Aircraft
MS 28010	Indicator, Pressure-Synchro, Single, 2-Inch Size

FSC 6930

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:03Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer.)

3. REQUIREMENTS

3.1 The requirements of MIL-I-82356 and this specification shall apply.

3.2 Design basis.- Each simulated aircraft torque pressure indicator shall conform to the applicable operational instrument specified in Table I of this specification. In each case the activating component shall be coupled to the vernier pointer with a 1:1 ratio and the vernier shall drive the main pointer with a ratio that will satisfy the applicable operational instrument display referenced in Table I.

3.3 Case.- With the exception of length, which shall not exceed the limits specified in the footnotes, cases shall be in accordance with Table I of this specification, as applicable.

3.4 Wiring .- Wiring shall be in accordance with the schematic referenced in Table I,as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-I-82356 and as specified in 4.2 through 4.3.

4.2 Tests.- The following tests shall be performed on each instrument in addition to tests specified in the test requirements of MIL-I-82356:

(a) Ratio of vernier pointer to main pointer

(b) Backlash of gears

(c) Forward and reverse operation

(d) Running smoothness of gears

(e) No jumping or hesitation of pointers.

4.2.1 Electrical tests.- Electrical tests for instruments referenced in Table I shall consist of checks on electrical zero and running performance. Test equipment shall consist of a transmitter synchro of a size equal to or larger than that employed in the instrument actuator to be tested. The test transmitter synchro shall be rotated by a precision dial where zero degrees has been made to correspond to the reference line at the transmitter's true electrical zero in accordance with MIL-S-20708.

4.2.1.1 Electrical zero.- Tests for electrical zero shall be conducted with the test transmitter synchro phase coordinated with the activated receiver synchro in accordance with MIL-S-20708. When the test transmitter

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:03Z
Check the source to verify that this is the current version before use.

synchro dial is at 0° ± 0.5 degrees, the direct driven pointer shall read its electrical zero ± 1.8 degrees.

4.2.1.2 Running tests.- Each receiver synchro, connected in proper phase compatibility as outlined in MIL-S-20708, shall be subjected to the following running tests using the test set-up specified in 4.2.1:

(a) Slow run - Each transmitter synchro shall be rotated clockwise and then counterclockwise at approximately 1 rpm for the full range of the indicator. The receiver synchro indicator pointer shall track smoothly without any discontinuities

(b) Fast run - The test of 4.2.1.2(a) shall be repeated in both directions at a rate of approximately full range in three seconds. The display pointers shall follow without any discontinuities and not lag the transmitter by more than 10 degres.

4.3 Activator ratio test.- The ratio between the vernier pointer and the main pointer for each display section shall be such as to satisfy the requirements of design of the operational indicator.

U. S. GOVERNMENT PRINTING OFFICE: 1969-341-519/A-6960

5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery requirements of MIL-I-82356 shall apply.

6. NOTES

6.1 The notes of MIL-I-82356 shall apply.

CUSTODIANS:
ARMY - AV
NAVY - TD
AIR FORCE - 70

USER ACTIVITIES:
ARMY - AV
NAVY - TD
AIR FORCE - 11, 70

PREPARING ACTIVITY:
NAVY - TD
Project Number 6930-0031

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:03Z
Check the source to verify that this is the current version before use.

TABLE I - SINGLE SYNCHRO DATA

DESIG-NATION	OPERATIONAL INSTRUMENT BEING SIMULATED	NOMINAL SIZE	*CASE TYPE	RATIO BETWEEN VERNIER AND MAIN POINTER IF APPLICABLE	SCHEMATIC WIRING MIL-I-82356 FIG. NO.
-1	MIL-I-7304	2 inch	MS 27572	40 to 1	4
-2	MIL-I-7307	2 inch	MS 28010	1 pointer	4

* Maximum case length of simulated instruments may exceed the case length specified in the specifications noted above. However, the maximum case length should not exceed 6 inches without prior approval from the procuring activity.

☆

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:03Z
Check the source to verify that this is the current version before use.